

1: [35 points total] You are interested in the effect of a binary treatment on an outcome of interest. To this end, you collect an iid sample of n individuals and assign them to the treatment or control group with equal probability. Let Y_i denote the outcome of the i th individual. It is assumed that $E[Y^2] < \infty$. Let D_i denote the treatment status of the i th individual, where $D_i = 1$ if the i th individual is treated and $D_i = 0$ if the i th individual is not treated. You assume the following model for Y_i :

$$Y_i = \alpha_i + \beta_i D_i$$

where (α_i, β_i) are independent of D_i . Note that α_i and β_i are random variables, and β_i is the effect of the treatment on the outcome. Furthermore the effect is "heterogeneous" in the sense that β_i differs across individuals.

1. [5 points] Explain why it might be reasonable to assume that (α_i, β_i) is independent of D_i .
2. [10 points] Consider the linear regression:

$$Y_i = \alpha + \beta D_i + \varepsilon_i$$

which you interpret as the best linear predictor of Y_i given D_i . Express α and β in terms of the distribution of (α_i, β_i) . In particular, show that $\beta = E[\beta_i]$

3. [5 points] Do you think ε_i is heteroskedastic or homoskedastic? Why?
(hint: try writing $\varepsilon_i = \alpha_i - E[\alpha_i] + (\beta_i - E[\beta_i]) D_i$ and investigating $\text{Var}(\varepsilon_i | D_i)$ for both possible values of D_i .)
4. [5 points] Define $X_i = (1, D_i)$ and define $\hat{\alpha}_n, \hat{\beta}_n$ as the OLS estimators of α and β given our sample. Recall from our usual asymptotic theory that we know:

$$\sqrt{n} \left((\hat{\alpha}_n, \hat{\beta}_n)' - (\alpha, \beta)' \right) \xrightarrow{d} N(0, \Omega)$$

Write down an expression for Ω as well as a consistent estimator of Ω . Call this estimator $\hat{\Omega}_n$. You do not need to show that $\hat{\Omega}_n$ is consistent.

5. [10 points] Suppose we are interested in obtaining a confidence interval around $\hat{\beta}_n$, but we do not care about $\hat{\alpha}_n$. Define $r' = (0, 1)'$ and show how to use this vector to derive a confidence interval for $\hat{\beta}_n$ at level $a \in (0, 1)$.